CHAPTER TWO

HOW SW & HW WORK TOGETHER

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Chapter Objectives

Upon completion of this chapter, the student will be able to:

- Classify buses in different ways
- Describe the common bus technologies
- Describe & explain the booting process
- Describe hardware resources in a PC
- Enter CMOS setup & perform configuration

Sub-topics

2.1 Introduction to buses
2.2 Technologies of buses
2.3 The booting process
2.4 HW resources in PC
2.5 BIOS/CMOS setup

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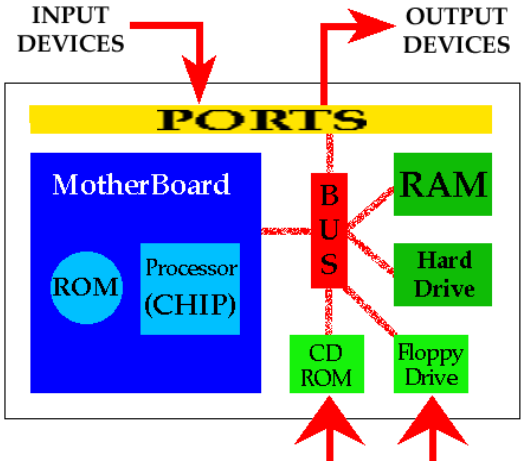
2.1 Introduction to buses



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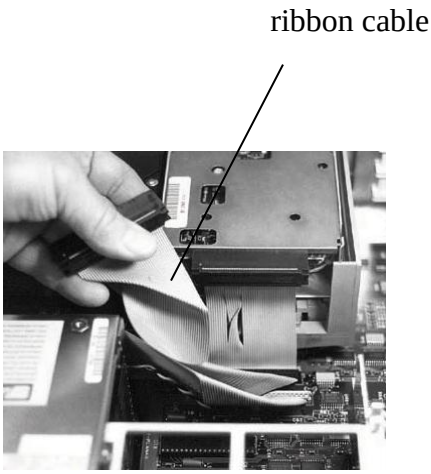


Buses are **electronic pathways** or **data channels** or **physical connections** along which binary bits (0s & 1s) pass e.g. metal tracks on PCBs, ribbon cables.

CPU uses buses to communicate, by sending **data**, **addresses** and **control signals** to components inside and outside the system unit.

Buses can be classified:

1. By location 2. By function 3. Broadly

Note:

All I/O devices communicate directly with each other and with main memory (RAM) via the different I/O buses, except the CPU.

Classification by location

1. Internal buses 2. External buses

Internal buses

These are system buses that connect the different internal components inside the system unit on the motherboard. They enable CPU, main memory (RAM) and all other components on the motherboard to communicate. For example, metal tracks (found on MB, expansion cards & slots).

External buses

These are peripheral buses used to expand the internal buses. They enable communication between CPU and external components (I/O devices & external storage devices). They can be parallel buses (e.g. ribbon cables) or serial buses (e.g. USB).They are generally slower than the internal buses (system buses). For example, USB ports and data cables for external hard disks.

Classification by function

1. Address buses 2. Data buses 3. Control buses

Address buses

Also known as memory buses; carry memory addresses from CPU to main memory (RAM) and to I/O devices (peripherals). They are unidirectional buses (one way traffic).The memory addresses are for memory locations that store data. The width (size) of the address bus corresponds to the number of memory locations that can be accessed. The CPU keeps track of which data is stored where, using memory addresses.

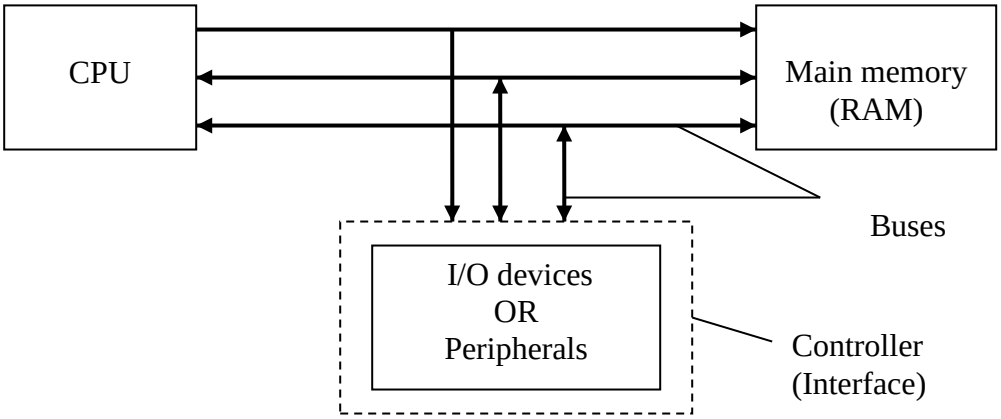
Data buses

They carry data from main memory (RAM) and I/O devices (peripherals) to the CPU for execution. They also carry data from CPU to main memory (RAM) and I/O devices (peripherals). This means they are bidirectional buses, but can only transmit in one direction at a time. Data is stored, manipulated and processed in main memory (RAM).

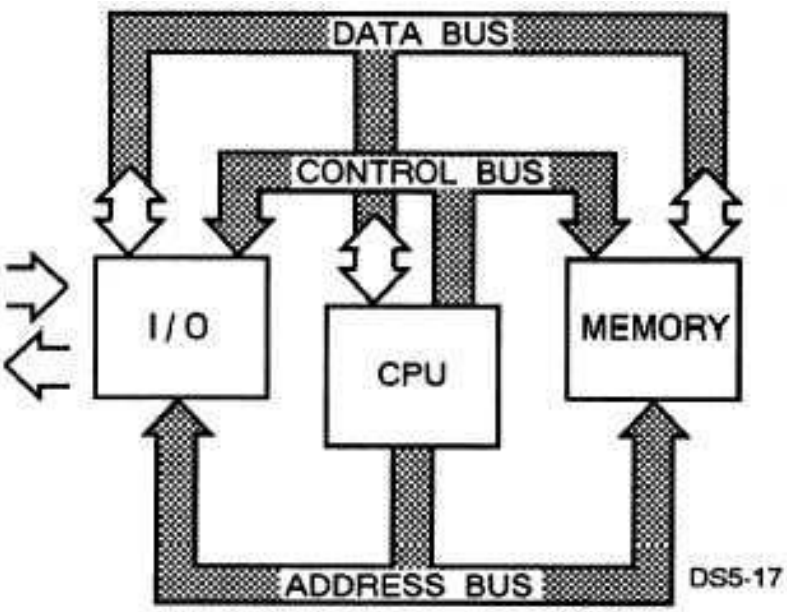
Control buses

Also known as command buses, carry instructions (control & coordination signals) from CPU (control unit) to main memory (RAM) and to I/O devices (peripherals). They also carry response signals from main memory (RAM) and I/O devices (peripherals) to CPU. This means they are bidirectional buses, but can only transmit in one direction at a time. The instructions (signals) include: read, write, interrupt, acknowledge.

See simplified diagrams below for the 3 types of buses above.



Top: address bus Middle: data buses Bottom: control buses



Classification broadly

1. System buses 2. Peripheral buses

System buses

Also known as local buses (older designs) or front-side buses (newer designs). These are internal buses between CPU and main memory (RAM). They transfer data at high speed using parallel communication e.g. 32-bit bus sends data over 32 wires simultaneously. Modern system bus is 64-bit and runs at 66, 100 or 133MHz.

Peripheral buses

These are pathways between CPU and peripherals (I/O devices) e.g. hard disks, RAM modules, video card, sound card, printer. They are extensions to system bus which ends in a chipset (a controller chip on the motherboard).Sometimes known as I/O buses, if they connect CPU to I/O devices. They transfer data at low speed by serial or parallel communication. They are either internal (e.g. ribbon cables) or external (e.g. ports).

How CPU performs its work

Generally it follows 3 stages: fetch – decode – execute.

CPU places device addresses on address bus (address lines). Controller (interface) decodes the addresses to determine the device. If the address is for the device the controller (interface) disables other devices. The command is decoded and executed by the CPU.

What to consider in designing a bus

1. Bus width
   High speed local bus should be wide to carry more bits.
   Difficult to design, because high speeds cause timing problems for shot distances.
2. Noise interference
   Noise interference should be minimized, because it causes signal distortion in metal tracks.

Bus width

Refers to the number of bits that can be transmitted at once (parallel communication).

Bus width is in multiples of 2 i.e. 2^2=4, 2^3=8, 2^4=16, 2^5=32 & 2^6=64.

For example, 32-wire ribbon cable can transmit 32 bits in parallel.

Bus width used for 2 things:

1. Data bus width i.e. size of the data bus
2. Processor word size i.e. data capacity of the microprocessor

For example, Intel microprocessors:

- Obsolete (1971-1979) had word sizes 4, 8 & 16.
- Still in use (1982-today) have word sizes 16, 32 & 64 (modern).

Data bus width and processor word size may or may not be equal.

The smaller of the above two determines the bus width of the PC.

For example:

- IBM PC/AT with bus width 16 (data bus 16 & word size 16).
- IBM PS/2 with bus width 32 (data bus 32 & word size 32).
- Apple Mac Plus with bus width 16 (data bus 16 & word size 32).

Bus speed

Also known as **data rate?** Or **frequency**, refers to number of times a group of bits can be sent per second. It is measured in Hz (hertz), but practical unit is MHz (mega hertz). It is the **number of bits per line per clock cycle** of CPU. Clock cycle (bus cycle) occurs every time data travels from memory to the CPU.

Most buses transmit 1 bit of data per line per clock cycle, and newer buses may move 2 bits of data per line per clock cycle e.g. AGP (Accelerated Graphics Port) for video cards.

Older buses may move 1 bit per line per 2 clock cycles e.g. ISA (Industry Standard Architecture).

Bus bandwidth

Also called **throughput**, refers to maximum **transfer speed** of bus. Refers to number of bits that can be sent to CPU simultaneously. It is measured in **bps (bits per second)**, but practical unit is mbps (megabytes per second).

Note:

Bus bandwidth (in bps) = bus width (in bits) x bus speed (in Hz)

For example, 16 bit bus width with frequency of 133 MHz:

Bus bandwidth = 16 x $133x10^6$ = $2128x10^6$ bps =266MBps

System RAM speed is controlled by:

1. Bus width (in bits) 2. Bus speed (in HZ frequency)

For example, theoretically:

- 100 MHz 32 bit bus sends 32 bits (4 bytes) to CPU at $100x10^6$ times per second.
- 66 MHz 16 bit bus sends 16 bits (2 bytes) to CPU at $66x10^6$ times per second.

Qn.

How much data does 100 MHz 32 bit send compared to 100 MHz 32 bit system through CPU?

2.2 Technologies of buses

These refer to technologies for expansion cards and expansion slots.

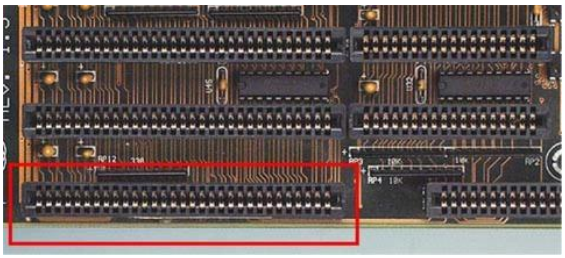
They include: **ISA**, **EISA**, **PCI**, **AGP**, **MCA** and **VL (VESA)**.

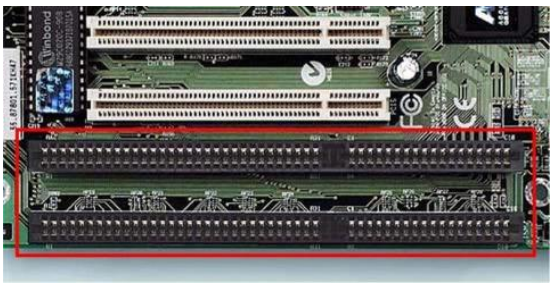
ISA & EISA are associated with old IBM PCs which are now obsolete.

Most modern PCs use PCI and AGP buses with higher speeds (MHz) & higher data rates (mbps). Other technologies not common do exist e.g. NuBus, TURBO channel, VMEbus, MULTIBUS, STD bus, CAN bus and FlexRay bus.

IBM had 9 competitors (*the Gang of Nine*) in bus technology i.e. AST Research, Compaq Computer, Epson, Hewlett-Packard, NEC, Olivetti, Tandy, WYSE, and Zenith Data Systems.

ISA bus





L: 8-bit ISA exp. slot R: 16-bit ISA exp.slot with alignment key

ISA (Industry Standard Architecture) developed by IBM in 1984.
Replaced 8-bit PC/XT bus (1980-82) at speed 4.77-6 MHz and data rate 4-6MBps. Physically identified as more than one (6 or 7) **black** expansion slots on the motherboard.
It was popular & most widely used for expansion cards and slots.
The 2 versions of ISA bus are:

- 8-bit ISA bus at speed 4.77MHz for PC/XT motherboard (1983).
- 16-bit ISA bus at speeds 8 or 10MHz (turbo) for PC/AT MB (1984).

Note:

- 8-bit ISA expansion slot with no alignment key, but 16-bit ISA has alignment key.
- 16-bit ISA bus synchronous with CPU at speed 8-10MHz & data rate 8MBps.

Disadvantages of ISA bus

1. Runs at lower speeds (8 or 10MHz), compared to e.g. PCI & AGP.
2. Expansion cards have power limitations i.e. 2 power devices on card.
3. Not plug & play i.e. configured using jumpers & dip switches.

EISA bus

EISA (Extended ISA) is 32-bit developed by Compaq in 1987. It provides backward compatibility to 16-bit ISA bus, by having 2 rows i.e. upper 16-bit & lower 32-bit. Runs at the same speed (8MHz) as 16-bit ISA bus, to be compatible. The design was exposed to all manufacturers i.e. compatible with other technologies (not proprietary**).**

Note:

- Compaq borrowed a lot of features from **MCA** and expanded on them.
- EISA bus never became popular in desktop PCs (unlike VESA & PCI) and can only be found in server PCs and higher-end PCs.
- 32-bit EISA bus for servers is synchronous with CPU 8MHz (max.) and data rate 32mbps.
- **SCSI Interface card** once used in a powerful network server is an example of EISA bus card.

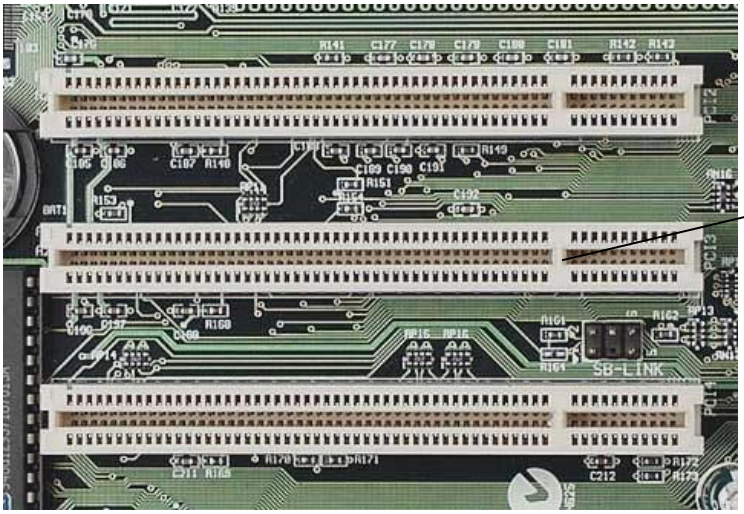
Advantages of EISA bus

1. Has higher bus width (32-bit) than ISA bus (16-bit).
2. Allows *backward compatibility* for 16-bit ISA bus.
3. Not proprietary i.e. compatible with IBM PCs and ISA expansion cards.
4. Allows more than one CPU to share the bus.

Disadvantage of EISA bus

More expensive to implement by MB manufacturers than ISA (though less than MCA).

PCI bus



PCI (Peripheral Component Interconnect) was developed by Intel in 1992. Physically identified as more than one (3 or 4) white or yellow expansion slots with alignment keys on the motherboard. It quickly evolved to replace ISA bus and is the most commonly used method for expansion cards. Intel created *PnP (plug & play)* standard and incorporated it into the design for PCI. With PnP, BIOS or OS determine HW resources (memory, I/O or interrupt) requirements automatically to avoid conflicts. It has been fully standardized by PCI SIG (PCI Special Interest Group). PCI provided a new way of connecting peripherals to both the main memory (RAM) and CPU.

PCI connects peripherals using:

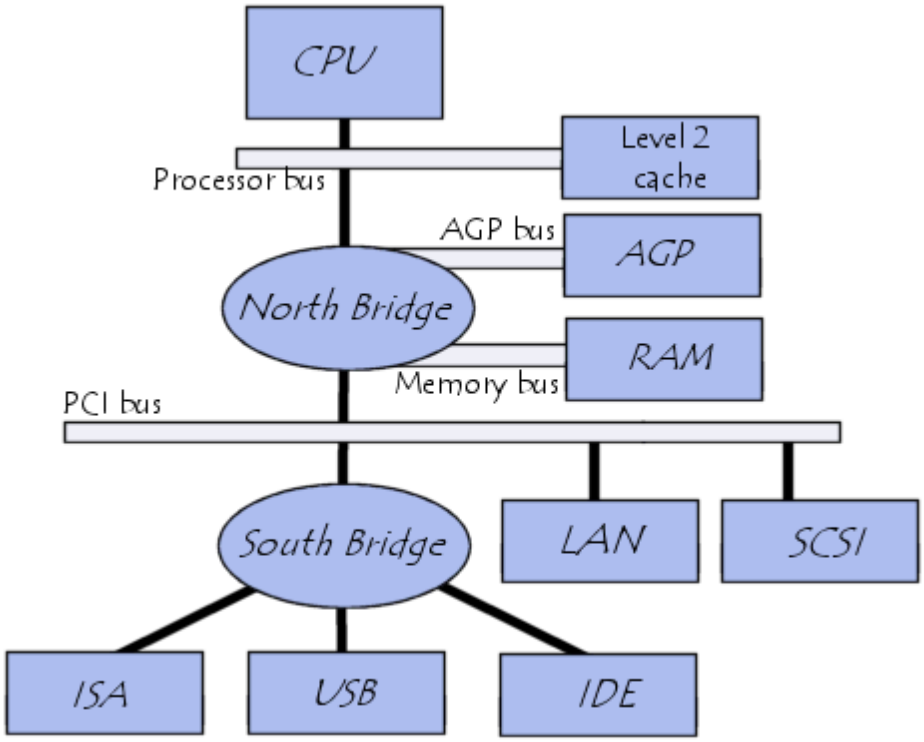
- PCI expansion slots on MB e.g. video card, sound card, NIC, TV card.
- Chipsets i.e. connecting North Bridge to South Bridge.
- On-board i.e. video, sound, IDE ports, SCSI controller and NIC.

See simplified diagram below showing PCI bus connections.

Versions of PCI bus are:

1. Original version PCI (64-bit, 33 MHz, up to 132MBps).

2. Later versions PCI-X (64-bit, 66 MHz to 133 MHz, up to 1066MBps).



Advantages of PCI bus

1. Runs at higher speed (33MHz), compared to ISA & EISA.
2. PCI standards include a full plug-and-play capability (automatic configuration).
3. Allows *backward compatibility* for older PCI & also ISA cards.

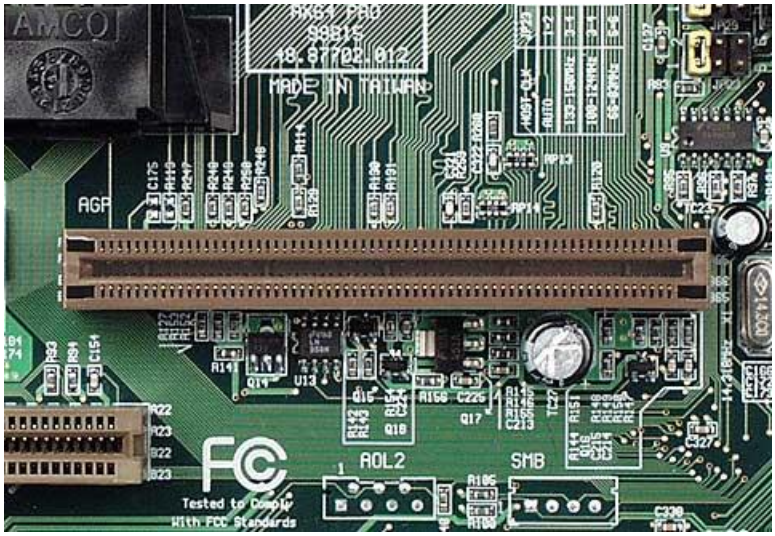
Plug and Play (PnP)

It means you can connect a device or insert expansion card into slot and it is automatically recognized and configured i.e. no need to set jumpers & dip switches (for old systems).

Disadvantages of PCI bus

1. Low data rate for newer video cards handling 3-D applications.
2. Bus with shared bandwidth (used to connect many peripherals).
3. Only 3 or 4 PCI expansion slots, compared to 6 or 7 ISA ones.

AGP bus



AGP (Accelerated Graphics Port) was developed by Intel in 1992.
Physically identified as only one *brown* expansion slot on motherboard. For specifically high speed video cards (graphics adapters). All versions are 32-bit with data rate 66MHz and bandwidth 133mbps (megabytes per second).
AGP bus uses RAM to generate 3-D images and video at high speed. Standard graphics port for all new PCs; most new PCs have a single AGP expansion slot. Uses AGP controller integrated on motherboard to enable video card communicate to CPU.
- Versions of AGP bus are: AGP x1, AGP x2, AGP x4 and AGP x8.

Note:
- PCI & AGP expansion slots are left empty for future upgrading.
- Before AGP bus, video cards were installed in PCI expansion slots.
- A second video card can be installed in a PCI expansion slot.

Advantages of AGP bus
1. Offers high speed & data rates (See table below for comparison).
2. Bus dedicated for only graphics cards (AGP bandwidth is not shared).

Note:
- PCIe (PCI-Express), is designed to replace the older PCI, PCI-X and AGP bus standards. (See advantages of PCI-Express bus).
- Many modern MBs have no AGP slots, since their chipsets have no AGP support.
- However, some MBs are produced with older chipsets that have AGP support.
- Many manufacturers of graphics cards continue to produce AGP cards for the disappearing AGP market.

Advantages of PCI Express bus

- Allows for higher data transfer rates
- Has more robust full-duplex support
- Supports other devices.

MCA bus

- **MCA (Micro Channel Architecture)** was a **proprietary** parallel bus was introduced by IBM in 1987 for use on new PS/2 computers.
- Proprietary means **incompatible** (not to standards) i.e. PCs with MCA bus used only MCA expansion cards.
- Because, MCA expansion cards were smaller and used different edge connectors.
- It was available in mainly 32-bit, but also 16-bit bus with clock speed 10MHz.
- The 16-bit was used to lower the cost in Intel-based machines e.g. PS/2.
- One of 6-bit slots had additional 20-pin **video extension connector** for graphics cards.
- The 32-bit slots had additional 8-pin matched **memory extension connector**.

Advantages of MCA bus over ISA bus

1. Higher bus speed (increased from 8 to 10MHz).
2. Bigger bus width (increased from 16 to 32-bit, though 16-bit existed).
3. Supported more interrupts (not fixed in HW as for ISA).
4. Supported more I/O device addresses ((not fixed in HW as for ISA)).
5. Supports plug and play (auto-configuration e.g. no jumper settings).

Disadvantages of MCA bus

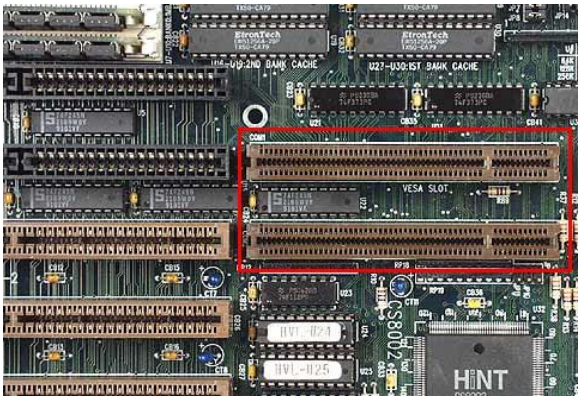
1. Proprietary (limited only to IBM PCs and but not compatible with ISA expansion cards.
2. Not as popular as ISA, because more expensive to implement by MB manufacturers.

Note:

- The MCA bus never became widely used and has since been phased out of desktop computers (disappeared with the introduction of PCI).
  Why? The small market for MCA made it very expensive.
- Very few MCA sound cards were ever produced compared to ISA and PCI sound cards.

VL or VESA (Video Electronics Standards Association's Local Bus)

Also known as VESA Local Bus (VLB) is a local bus type i.e. runs at the same speed as CPU. PCI local bus replaced VL or VESA local bus.



Advantages of VL

1. Commonly used for video to gain the advantage of the high speed of transfer of CPU.
2. Supports both 16-bit (allows backward compatibility for 16-bit ISA bus) and its own 32-bits (has extra expansion slot).

2.3 The booting process

Booting means to load the OS (Operating System) and utility programs into main memory (RAM).

For example, Windows OS is loaded when icons on the desktop form.

A small program called **boot program** is the startup program of PC.

Boot is short form for **bootstrap** (used for old computer systems).

There are 2 types of booting:

1. Cold booting 2. Warm booting

Cold booting

This is the process of switching on the PC from the mains wall socket. **POST** (Power on Self-Test) routines run first, then later OS is loaded into main memory (RAM).

Warm booting

This is the process of restarting the PC when it is already switched on. The PC can be started again from stand-by or hibernating modes. This type skips POST and immediately OS is loaded into RAM. Warm booting is used as a **troubleshooting technique** e.g. when a program is not responding (taking long).

It is done by one of the following:

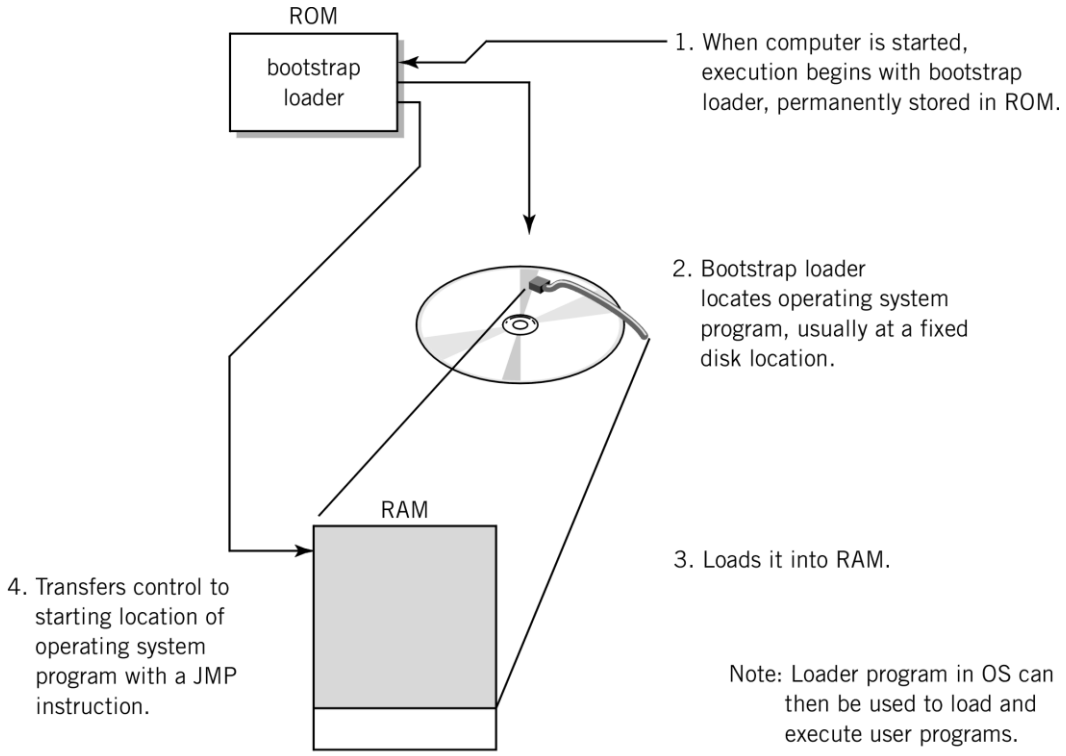
- Reset button on PC
- Windows restart command i.e. Start>Shutdown>Restart.
- Combination of keys i.e. Ctrl+Alt+Del.

How can you decrease boot time?

POST (Power on Self-Test)

POST starts the procedure (program) and performs the following tasks:

1. Tests RAM
2. Locates and checks devices e.g. keyboard, mouse, drives.
3. Searches drives (FD, HD and CD-ROM) for bootable devices.
4. Loads boot sector from a bootable device into RAM.
5. Transfers control of computer to OS to take over.



Note:

- The **boot loader** (e.g. ntldr in Windows OS) resides in **boot sector** of bootable device.

- The boot loader takes control of the computer from the POST process.
- It is also responsible for locating and loading the rest of OS files.
- Error may be reported e.g. if no bootable device is missing, OS is corrupted by viruses.

ROM BIOS chips

ROM is *Read Only Memory*, means not updatable (originally) and **BIOS** is *Basic Input Output System*. **ROM BIOS** chips are embedded on motherboard during manufacturing.
Also called *System BIOS*, is **firmware** (store fixed data) i.e. a small internal fixed programs used to control devices in a PC. They hold boot program and other utility programs (e.g. drivers which control devices). **Boot sequence** is the order BIOS follows to check for the OS to load in RAM.
Recommended order is:

1. CD-Rom drive (CD or DVD)
2. Drive C -hard disk drive.

Types of BIOS and their manufacturers are:

1. Phoenix BIOS by Technologies Ltd.
2. AMI BIOS by American Megatrends Inc.
3. MR BIOS by Micro ID Research

Check other types on the Internet.

Interfaces

They are circuitry needed to connect I/O devices/peripherals to buses.
They are used to manage transfer of data between devices/peripherals and CPU.
Examples of interfaces are:

- Ports on expansion cards and I/O devices.
- Connectors of ribbon cables and hard disk drive.

Controllers

They are circuitry or chips with **program logic** (intelligence) used to operate and control a device.
Initiates commands and senses status of a device.
Controllers are found on motherboards or on expansion cards or housed inside devices, depending on technology.
Devices of the same type can share the same controller e.g. drives (hard disk drive, CD-ROM drive).
Examples of controllers are:

- IRQ, DMA, USB, IDE controllers on the motherboard.
- AGP or CRT, SCSI controllers on expansion cards.
- Printer controller on printer.

See hard disk drive technologies later-Chap 6 e.g. IDE, SATA & SCSI.
See also IRQ and DMA controllers below.
Note:
Printer controller controls paper motion, print timing and selection of printing characters.

Why are interfaces and controllers needed in a computer system?
CPU differs from devices in speed and formats used in communication.
Both CPU and RAM have high speed (synchronized), but devices are very slow.

Summary of differences

CPU	***Devices***
High speed	Low speed
Uses digital (binary format)	Use different formats e.g. sound (audio), video (graphics)
Electronic in nature	Electrical or mechanical

2.4 HW resources in PC
HW resources allow I/O devices (peripherals) and CPU to communicate with each other.
HW resources include:
1. IRQs
2. Memory addresses
3. I/O addresses
4. DMAs

The device is first assigned HW resources before CPU can communicate to it.
See configuration data in BIOS/CMOS setup.
To view HW resources using Windows OS
Go to System Information>Hardware Resources. How?

IRQ
IRQ stands for **Interrupt Request** and simply known as **interrupt**.
There are 2 types of interrupts:
1. Hardware interrupts -bus feature (adapter or unit on I/O bus).
2. Software interrupts -used to call BIOS routines.

We are concerned with hardware interrupts in this course unit.

Hardware interrupts
The hardware interrupt allows a device to signal the CPU for immediate attention.
The device signals CPU to suspend whatever it is doing and attend to it.
The signal is a voltage applied to one of the lines (wires) in the bus.
Physically, IRQ is a wire (line) on the bus.

CPU acknowledges the signal, suspends **process** (running program) and attends to the device (branches to interrupt service routine).
The service routine processes the interrupt and when complete, returns control to the suspended **process** (running program).
Examples of interrupts are:

- A hard disk drive ready to receive or send data.
- A printer running out of paper and needs to load.
- Interrupt generated by a keystroke or a mouse click.

IRQ signals run along IRQ lines (wires) to the IRQ controller on the MB.
IRQ lines (wires) connect to all expansion slots, making the expansion cards (adapters) to communicate with CPU.
IRQ controller assigns priorities to incoming IRQs to be sent to CPU.
IRQ lines (wires) are also known as **IRQ channels**.
When you install expansion card in a vacant expansion slot, one of the IRQ channels is assigned to it automatically (plug and play).
Original PC/XT bus had 8 IRQ channels, while ISA bus has 15 IRQ channels.
But newer PCs have a total 16 IRQ channels (or more), of these 5 IRQ channels are internal and cannot be used with I/O cards.
IRQs can be assigned manually in BIOS/CMOS set up or by plug and play (auto-configuration).

Some standard IRQs

IRQ 0	System Timer (High Precision Event Timer)
IRQ 1	Keyboard
IRQ 3	COM2 & COM4 (Serial ports)
IRQ 4	COM1 & COM3 (Serial ports)
IRQ 5	LPT2 (Parallel port)
IRQ 7	LPT1 (Parallel port)
IRQ 8	RTC (Real Time Clock)
IRQ 11	Controller
IRQ 13	Numeric data processor
IRQ 14	Primary IDE
IRQ 15	Secondary IDE
IRQ 16	NIC (Fast Ethernet NIC)

Note:

1. Missing IRQs e.g. 2-7, 10, 12, 17, 20-22
   These IRQ numbers have been reserved and are not available for Windows OS to issue.
2. Usually one device per IRQ
   Assign IRQ to a new device when configuring & physical installation.

Plug & Play (auto-configuration) can sometimes do it smoothly.

3. IRQ sharing e.g. 16 & 23
   Two devices can share the same IRQ channel if they cannot request at the same time e.g. IRQ 16 shared by NIC and USB controller.
   PCI bus technology has helped IRQ channels to be shared easily.
4. IRQ conflict
   Can be handled by changing IRQ number (in CMOS setup) for device and assigning it to another device e.g. IRQ of sound card given to modem.
5. Free IRQs
   Some IRQ channels are normally available for SCSI, NIC and sound card.
   Because IRQs are limited, BIOS disables some ports not in use e.g. serial & parallel port disabled (USB ports care enabled).

Memory addresses

Memory addresses are addresses for memory locations called **cells**.
A memory address is assigned to each device and gives the memory location of the required subroutine the CPU needs to service the device.
These subroutines are usually found in the ROM BIOS or other ROM chips.
CPU specifies memory addresses of memory locations and I/O devices.
Addresses are transmitted along address buses (lines) using appropriate interface.

I/O addresses

These are unique memory addresses where information about each I/O device is kept.
CPU sends information to or reads information from the I/O address.
Some important components (e.g. hard disk controller, video card) have I/O addresses reserved for them.
While others (e.g. printers, scanners) need to have I/O addresses assigned (during configuration).
I/O addresses can be assigned manually in BIOS/CMOS set up or by plug and play (auto-configuration).

Note:
Both memory and I/O addresses are bit combinations forming address space.
For example:
2 bits give $2^2 = 4$ addresses, with address space = 00, 01, 10, 11
General formula: n bits give 2^n addresses.
Both memory and I/O addresses are usually expressed in hexadecimal (base 16), with characters = 0, 1, 2, 3,… 9, A, B, C, D, E, F

Some memory addresses

0xFDF00000-0xFDF7FFFF Chipset OK <See Chipset later>
0xFEC00000-0xFEC00FFF System board OK <See motherboard>
0xA0000-0xBFFFF PCI bus OK <See Bus technologies >

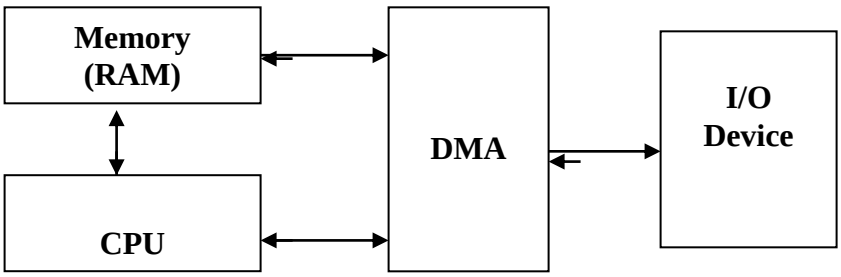
Some I/O addresses

0x00000000-0x00000CF7	DMA controller <See DMA below>
0x00000060-0x00000060	Keyboard
0x000001F0-0x000001F7	Primary IDE Channel <See hard disk>
0x0000DE00-0x0000DEFF	Fast Ethernet NIC <See expansion cards>
0x0000ED00-0x0000ED07	PCI Parallel port (LPT3)
0x0000F500-0x0000F51F	USB Universal Host Controller

Note:
- Some devices need more memory locations (addresses).
- Some devices have a set of addresses used as for different functions.

DMA

DMA stands for **Direct Memory Access**, which is a bus feature that allows devices to transfer data between main memory (RAM) and I/O devices without CPU intervention.
The devices bypass CPU and write their information directly into main memory (RAM). Valuable CPU time is spared for other devices or tasks.
This is another **I/O interfacing technique** on addition to IRQ.
DMA controller (found on MB) communicates to CPU on behalf of the I/O device ie it is designed to service one or more I/O devices.
CPU only initiates DMA by sending DMA address located in memory.
When transfer of data is complete DMA controller sends an IRQ to CPU.
DMAs are used by high speed devices, where large amounts of data are transferred eg magnetic disks (hard disks, tape drives).



DMA channel is a service interface and standard PCs have two DMA controllers (DMA1 & DMA2) and a total of 8 channels.

1. DMA1 has Channels 0, 1, 2 & 3 –use 1 byte (8 bits) per transfer.
2. DMA2 has Channels 4, 5, 6 & 7 –use 2 bytes (16 bits) per transfer.

DMA channels can be assigned manually in BIOS/CMOS set up or by plug and play (auto-configuration).
See BIOS/CMOS setup below.

Default DMA channels

0	DRAM refresh
1	Sound card
2	Floppy disk drive
3	ECP parallel port
4	DMA controller
5	Open (sound card)
6	Open
7	ISA hard disk controller

Note:

1. Two devices should not share same DMA channel
   If two devices are assigned same DMA channel, both will not work.
   Each type of bus has different DMA channels available.
2. **Clock signals** are measured in MHz (mega hertz) or GHz (giga hertz).
3. Internal click determines CPU speed, while bus clock determines the bus speed.
4. **Bus mastering** is a bus feature where some expansion buses take direct control of the external data bus and perform tasks without CPU intervention.
5. **Bus mastering** is much faster than **DMA** and therefore tasks are accomplished faster.

2.5 BIOS/CMOS setup

CMOS RAM chip

It is a chip found on the motherboard used to store startup configuration data (information).
Obsolete PCs e.g. PC-XT used DIP (Dual Inline Package) switches and jumpers for configuration.
Older PCs e.g. PC-AT by IBM used Ni-Cad battery for CMOS RAM to hold some of the system's advanced configuration information.
Modern PCs use a low power source i.e. a non-replaceable lithium cell on the motherboard that gives 1 micro amp current and can last 10 years.

Qn. What is the difference between ROM BIOS and CMOS RAM?

CMOS RAM (volatile) is *Complementary Metal Oxide Semiconductor*, while **ROM BIOS** (non-volatile) is *Basic Input Output System* (also known as System BIOS). *<See ROM BIOS above>*

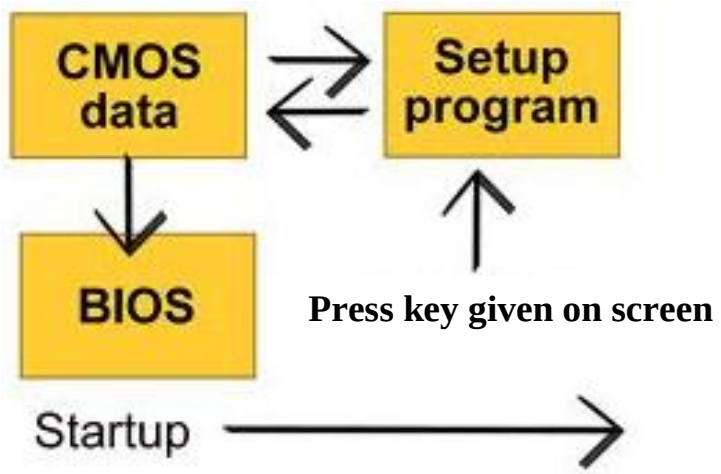
BIOS and CMOS are 2 different microchips on the motherboard, though related. How are they related? CMOS RAM do not store programs, but instead it is where the BIOS stores system settings or configuration data (information) for startup e.g. date and time, disk information.

ROM BIOS contains a program for booting a PC and is only modified or updated with *BIOS updates*.

CMOS RAM is powered by a CMOS battery and is modified (changed) by entering the CMOS setup.

Note:

- Although the setup is often referred to as the BIOS / CMOS setup, it is more appropriate to use CMOS setup.
- Manufacturers of motherboards buy BIOS from BIOS manufacturers (eg Phoenix BIOS, AMI BIOS) and then may make a few changes.



BIOS updates

Computer or motherboard manufacturer periodically updates the BIOS to e.g. fix bugs (errors), add compatibility with new devices, to speed up boot time.

The BIOS updates are available at the manufacturer's website.

How to update BIOS

1. Identify your current BIOS version.
2. Check your PC/motherboard manufacturer's Website for BIOS updates.
3. Read the included documentation.
4. Update your BIOS.

Precaution!

If you make a mistake in the BIOS update process, you risk rendering your PC unbootable.

Configuration data (information)

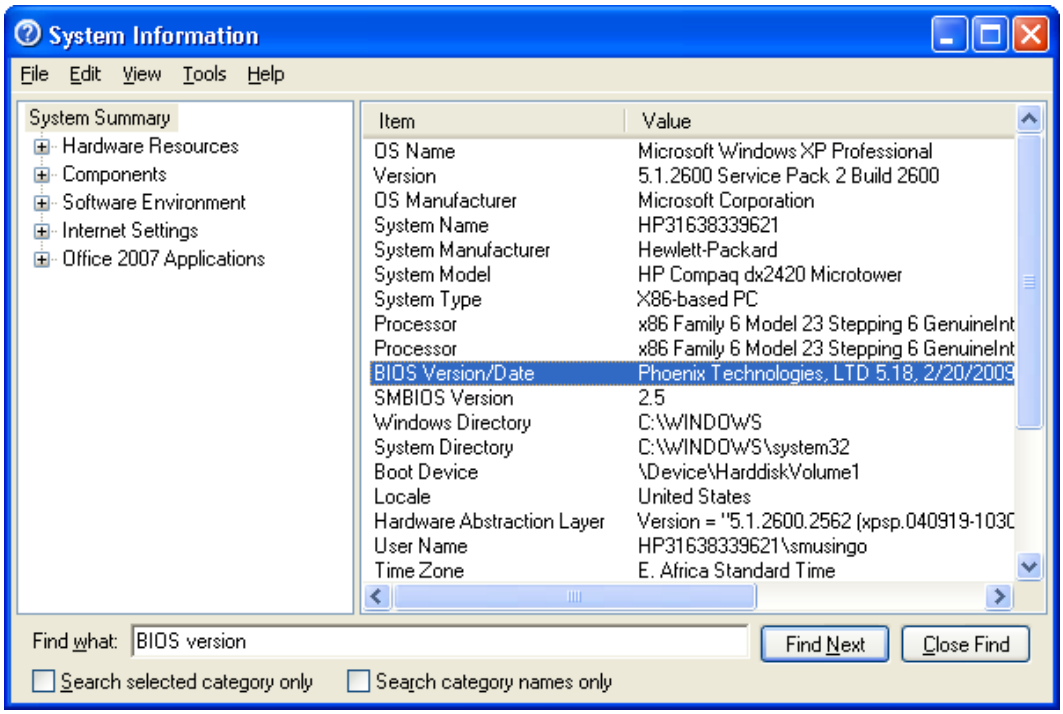
CMOS setups (system settings) may differ, depending on the manufacturer of the BIOS version e.g. Phoenix BIOS or AMI BIOS.

Basically CMOS setup includes the following:

1. Date and time
2. CPU information e.g. speed in MHz or GHz
3. Drive types e.g. hard disk, CD-ROM, flash disk
4. Memory information e.g. RAM capacity in MB or GB
5. Port information e.g. serial and parallel ports
6. HW resources available e.g. IRQ, DMA, I/O addresses
7. Power saving information
8. System security e.g. chassis intrusion

Qn. How do you to find BIOS version of a PC using Windows OS?

1. Go to System Information. How?
   Start>All Programs>Accessories>System Tools>System Information.
2. Type **msinfo32** into the search bar >Find Next.
   e.g. BIOS Version/Date Phoenix Technologies, LTD 5.18, 2/20/2009



Storage of configuration data

The 2 types of CMOS screens are:

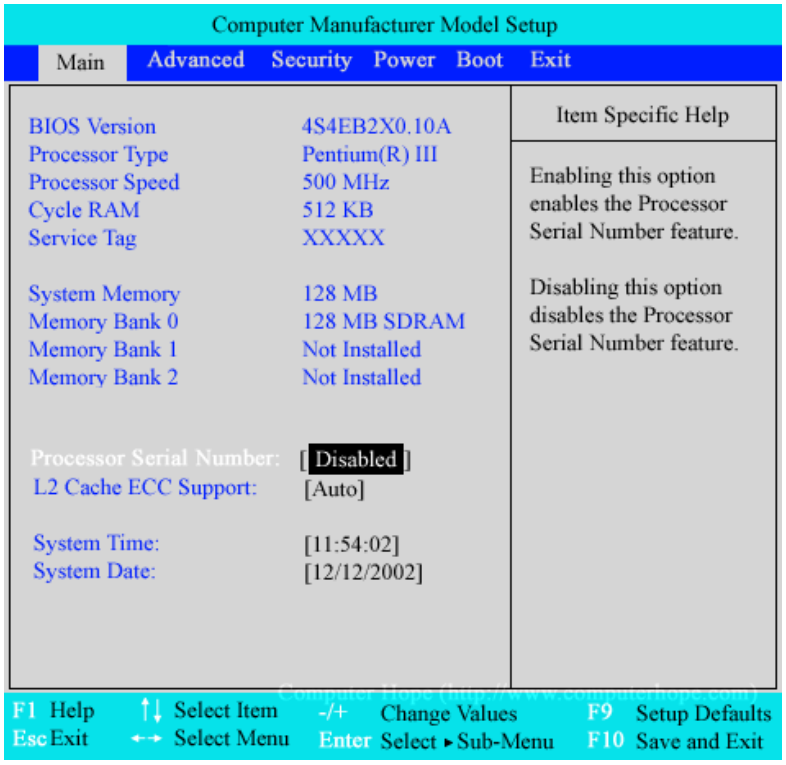
1. Text based –most common
2. Windows environment –for new PCs

To run a CMOS setup program

1. Press key shown on screen at setup (just before booting) Normally used are: Del, Ctrl, ESC, F2, F10
2. This program is independent of OS (loaded or not loaded) and no need to have hard disk drive.
3. It depends on the manufacturer of BIOS, but not manufacturer of PC. BIOS manufacturer and version number are displayed.

To change CMOS setup (system settings)

1. Select item and use keyboard keys e.g. Setup, Exit, Save and Exit.
   For example select:
   Time & Date (Set new time & date if incorrect)
   Boot sequence (Use recommended order, hard disk drive not 1st)
   Resources available (Assign IRQs to devices or rectify any conflicts).
2. Exit and check whether information set is still there.



Note:
System settings are the only way of getting information about what options are installed. If not set accurately an ERROR occurs.
Suspect configuration problems if a PC fails to operate immediately after a new device has been installed.
For example, if the hard disk drive information is incorrect it might not be accessible.
When this happens, it is advisable to reset the CMOS setup so that is properly defined in the CMOS registry.

Older PCs used hardware jumpers or manually entering CMOS setup.
Modern PCs use *plug and play* (PnP) to configure automatically e.g. PCI expansion cards.
PnP was discovered by a combination of Microsoft and Intel programmers and include in newer Windows OS. During the startup process, the *PnP BIOS* looks through the system for installed devices. Several devices contain their own flash BIOS (updateable ROM BIOS)
E.g. SCSI controllers for hard disks and video cards.

Note:
- It is important for the core information on a CMOS RAM chip to be correct.
- If you change any of the related hardware, the CMOS must be updated to reflect those changes.
- If the CMOS loses power from its battery, it will lose its data. The next time the system is started, the setup program will revert to its default settings.
- It is a good idea to write down the primary system settings (like hard drive parameters) and tape them inside the case for reference.

Some errors due to lost or corrupt CMOS data
1. CMOS configuration mismatch
2. CMOS date/time not set
3. No boot device available
4. CMOS battery state low
5. Cannot locate hard disk drive or floppy disk drive

ESD (Electrostatic Discharge)
This is static electricity that can damage ICs (Integrated Circuits), also known as microchips or simply chips e.g. ROM BIOS and CMOS RAM. ICs are made from semiconductor (e.g. silicon) and insulating materials (e.g. silicon dioxide). Either of these materials can suffer permanent damage when subjected to high voltages.
ESD is mainly caused by 2 ways:
1. Friction
2. Induction

Friction

For example rubbing your body on the chair or walking across the floor or removing a carpet from the floor.

A spark may be produced in dry conditions when touching a doorknob after walking across a carpeted floor.

Induction

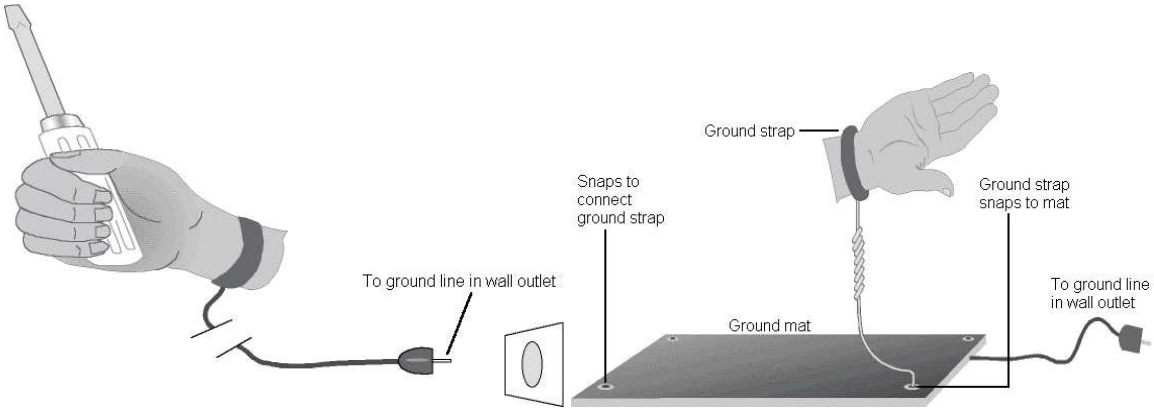
For example unwrapping a PCB (Printed Circuit Board) from a plastic container.

Precautions against ESD

1. Avoid touching pins / leads of microchips, because hands may have accumulated charges (ESD).
2. Discharge yourself (grounding) before working on any of the electronic components on the printed circuit board.

Why?

To ensure zero potential between your body and the PC.

How?

1. Touch unpainted metallic part of PC chassis or power supply.
2. Use ESD wrist strap or grounding mat or table.
3. Stand all the time, since sitting creates ESD on the chair.
4. Avoid putting on woolen clothes, because they create ESD.